

Outset · AI in Healthcare · Day 3 · Group 6

Does estrogen keep the mind sharp?

A twenty-year lesson in confounding, and why a survey can hint but only an experiment can prove

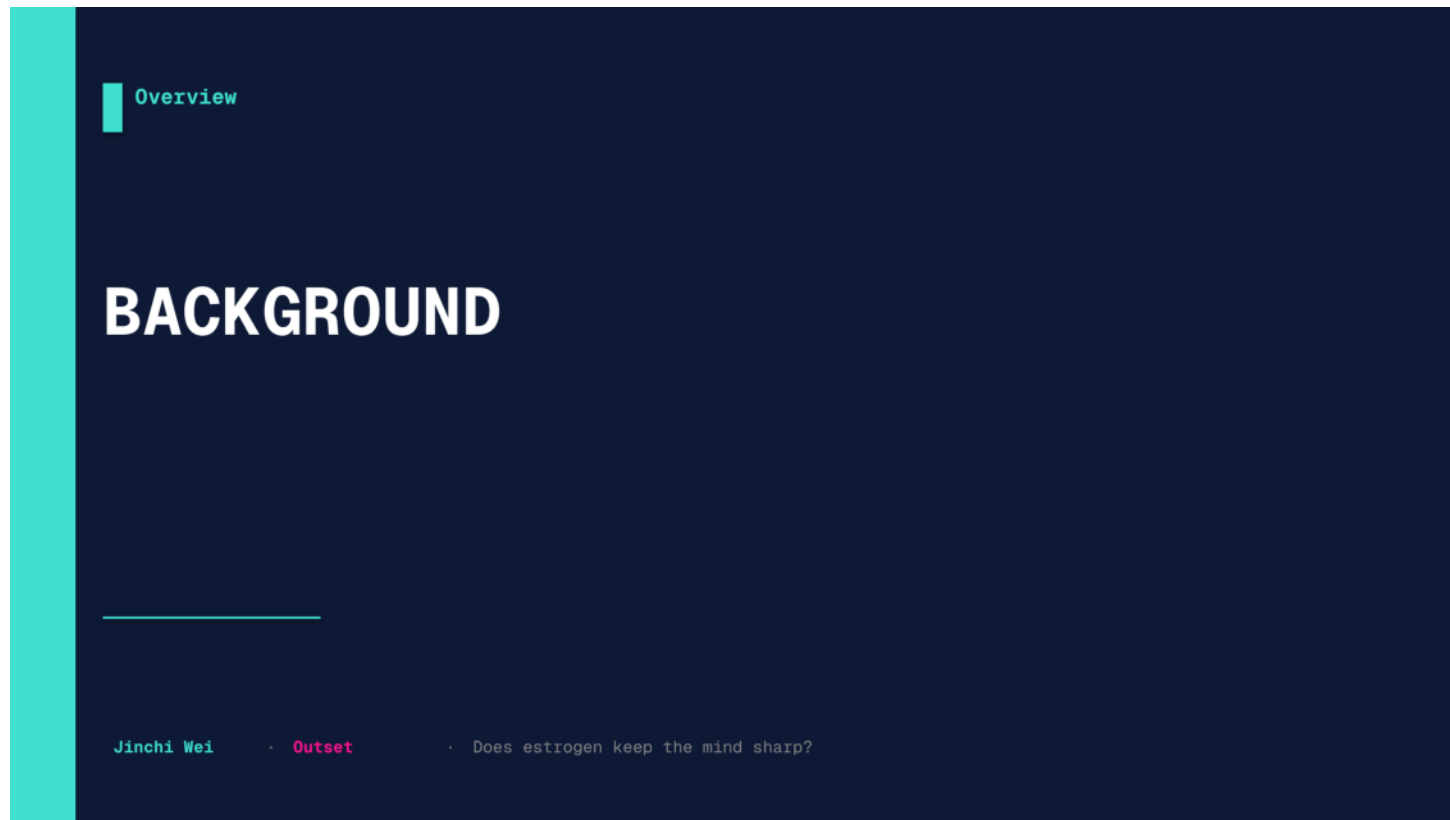
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Outset

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Slide 1 -



Slide 2 – Background

This opens the Background section. Everything ahead sets up the central mystery: how could years of just watching women and a single coin-flip trial reach opposite conclusions about the very same hormones? For now keep the audience curious, not yet informed. Transition: start with the twenty-year detective story.

A twenty-year detective story

THE EXCITING CLUE (1996-2002)

Studies that simply WATCHED women found that estrogen users developed Alzheimer's less often, and later, than women who never took it [1, 2]. For years this looked like solid proof that the pills protected the aging brain.

THE PLOT TWIST (2003)

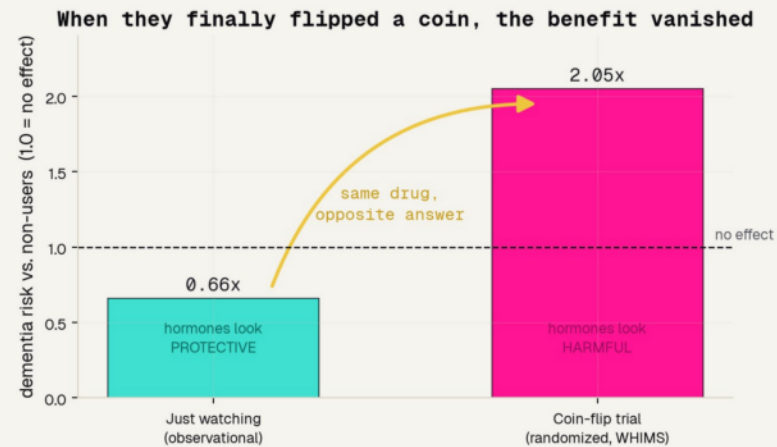
Then the WHIMS trial let a COIN FLIP -- not the woman herself -- decide who got hormones. The memory benefit vanished, and dementia risk actually ROSE in women aged 65 and older [3, 4].

Same drug, opposite answer. The rest of this talk is one question: HOW?

Slide 3 – A twenty-year detective story

Open like a mystery, because it was one. For twenty years, observational studies -- just watching women, not assigning anything -- kept finding that estrogen users had sharper memory and less Alzheimer's [1, 2]. Doctors nearly started prescribing hormones for the brain. Then in 2003 the WHIMS randomized trial flipped a coin to decide who got hormones, and the benefit didn't just vanish, it reversed: dementia risk actually rose [3, 4]. Same drug, opposite answer. Set up the whole talk as: how can two honest studies disagree this badly? Transition: here it is in one picture.

When they flipped a coin, the benefit vanished



adapted from Tang 1996 / Zandi 2002 (observational) vs. Shumaker 2003 WHIMS (randomized)

Just watching (turquoise) said hormones PROTECT; the coin-flip trial (pink) said they HARM -- the study design, not the drug, decided the answer.

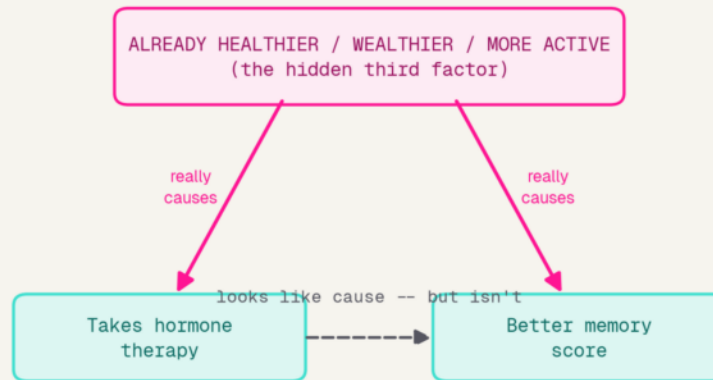
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Slide 4 – When they flipped a coin, the benefit vanished

This is our most important slide -- read it slowly. Left bar, 'just watching': hormone users looked protected, risk about two-thirds of non-users. Right bar, the coin-flip trial: risk more than doubled. The amber arrow is the sign flip. The dashed line is 'no effect.' The point for a 15-year-old: the STUDY DESIGN, not the drug, decided the answer. Watching can lie; a coin flip cannot. Transition: so why does watching lie here?

Healthy-user bias: a hidden third factor

Healthy-user bias: a hidden factor fakes the link



concept -- adapted from Wharton 2009 / Vandenbroucke 2009

Who chose the pill?

Users were richer, more active, and saw doctors more often to begin with [5].

The hidden factor

That same head start drives BOTH the drug use AND the good memory.

So the bottom link

only LOOKS like cause. It is a mirage -- confounding, a.k.a. healthy-user bias [6].

Slide 5 – Healthy-user bias: a hidden third factor

Now name the villain: confounding, or 'healthy-user bias.' The women who chose hormone therapy were, on average, richer, more active, and saw doctors more [5]. That hidden head start -- the pink box at the top -- pushes down on BOTH boxes: it makes them more likely to take the pill AND more likely to have good memory. So the dashed bottom arrow, 'takes pill → better memory,' only LOOKS like cause. It's a mirage created by the third factor [6]. This one diagram is the whole statistical idea of the talk. Transition: you already know this trap from summer.

Ice cream and shark attacks

On hot days, ice-cream sales and shark attacks BOTH rise together. Does ice cream summon sharks?

THE ANALOGY

No -- the hot SUMMER WEATHER drives both at once: heat sells more ice cream AND sends more people into the ocean to swim. Neither one causes the other.

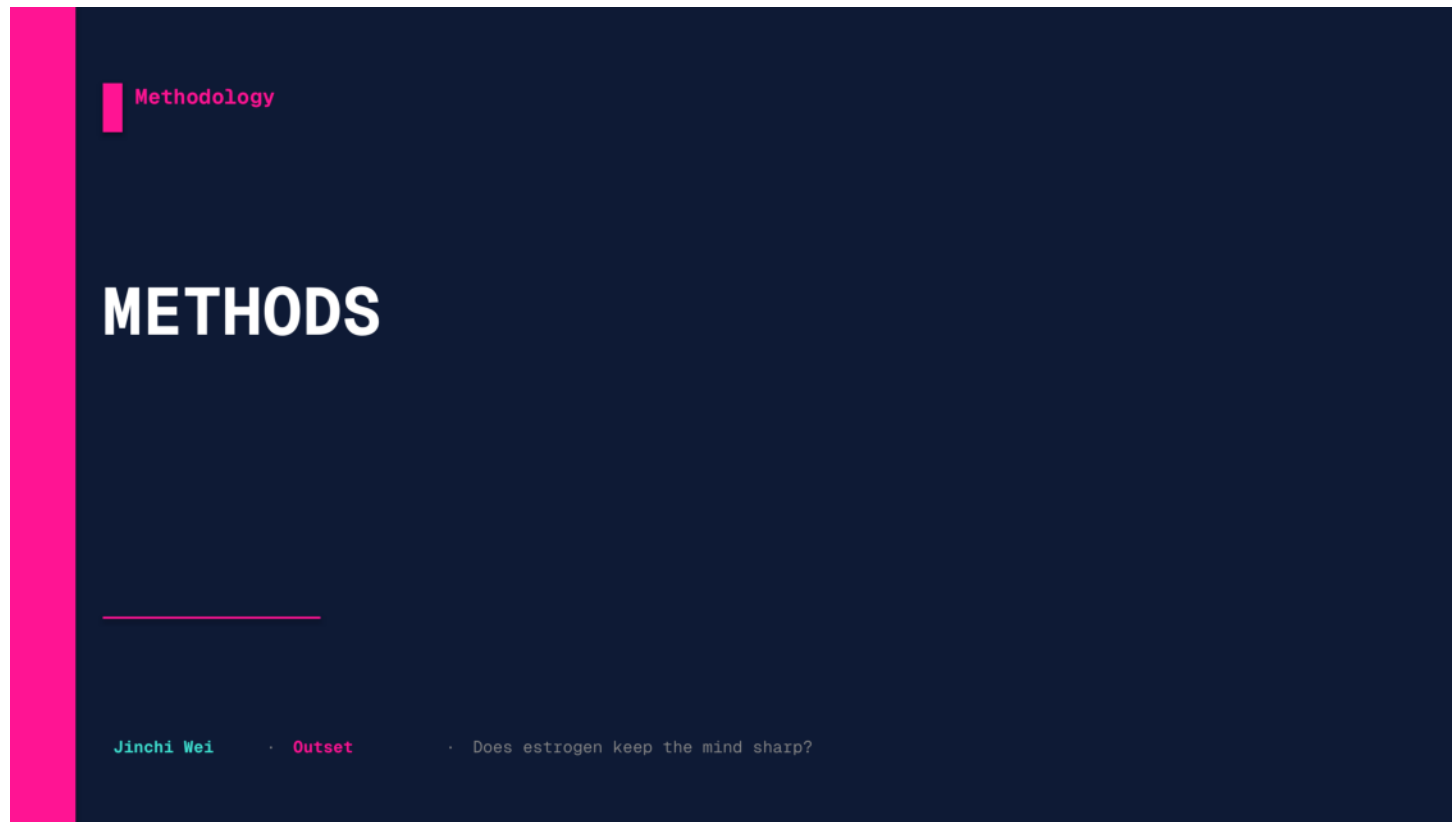
OUR VERSION

Estrogen use and good memory also rise together. But being HEALTHIER and better-off to start with can drive both -- so the pill itself may summon nothing.

When two things move together, hunt for the hidden third factor before you believe one caused the other.

Slide 6 – Ice cream and shark attacks

Make it click with the classic. Ice-cream sales and shark attacks rise together every summer -- but ice cream doesn't summon sharks; hot weather drives both. Map it: estrogen use is the ice cream, good memory is the shark attacks, and 'being healthier to start with' is the summer heat. The takeaway line is the transferable habit: when two things move together, hunt for the hidden third factor before you believe one caused the other. Transition: now to our own toy data.



Slide 7 – Methods

This opens the Methods section. We move from the big idea -- confounding -- to the concrete: the actual survey data we used and the one honest question a survey can answer. Plain-language framing: here is exactly what we looked at, and what we can fairly ask of it. Transition: meet the dataset.

Our data: NHANES women 60 and older

739 WOMEN

all aged 60 or older, drawn from NHANES 2013-14 -- a real US government health survey, not data we made up.

USED ESTROGEN?

whether she ever took female hormones -- a simple yes or no, self-reported by each woman.

HER BACKGROUND

her age, education, and family income -- the confounders we have to account for.

MEMORY SCORE

a digit-symbol attention test; higher = sharper. Our model tries to flag the weaker-scoring half.

Slide 8 – Our data: NHANES women 60 and older

Quick tour of the dataset so the results land. 739 women, all 60+, from NHANES -- a real US government health survey, not something we made up. For each woman we know: did she ever use estrogen; her age, education, and income (the possible confounders); and a memory/attention test score, where higher is sharper. Our model tries to predict who lands in the weaker-scoring half. Keep it light -- they don't need every column, just the shape. Transition: what can we actually ask of survey data?

The one question we can actually ask

We cannot randomly ASSIGN estrogen in a survey -- so we can never prove cause here. But we can still ask a narrower, honest question:

Does the estrogen effect SURVIVE once we account for age, education, and income?

CRUDE

Estrogen's effect measured all by itself -- ignoring everything else about the woman.

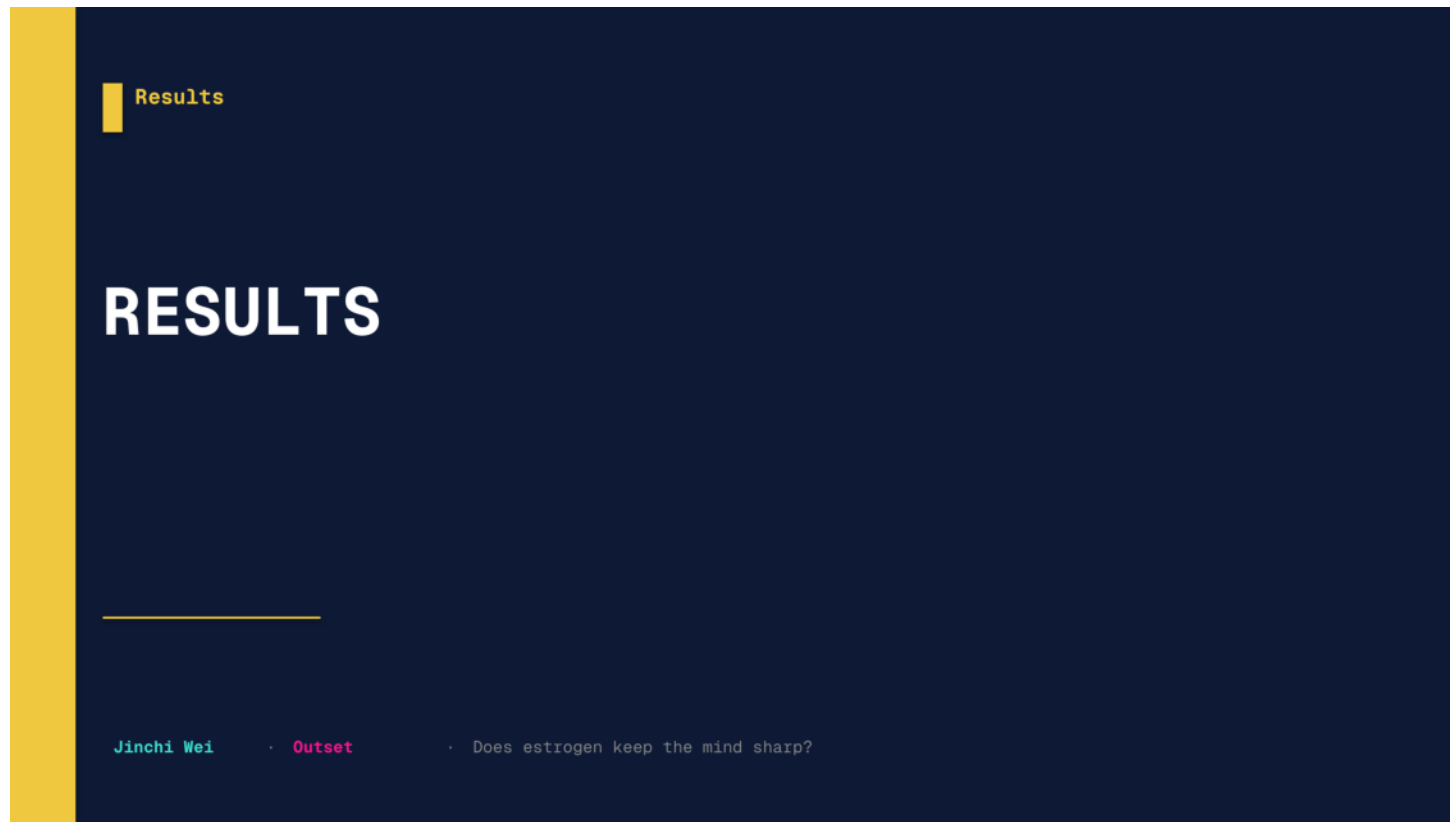
ADJUSTED

Estrogen's effect AFTER we account for her age, education, and income.

If the effect SHRINKS from crude to adjusted, the exciting headline was mostly confounding -- not real cause.

Slide 9 – The one question we can actually ask

Set expectations honestly. We can't randomly assign estrogen in a survey -- so we can't prove cause. But we CAN ask the narrower question in the box: does the estrogen effect survive once we account for age, education, and income? We'll measure estrogen's effect two ways -- crude (alone) and adjusted (after the confounders). If it shrinks, the crude headline was mostly confounding. That crude-vs-adjusted move is the entire method. Transition: start with the naive, tempting look.

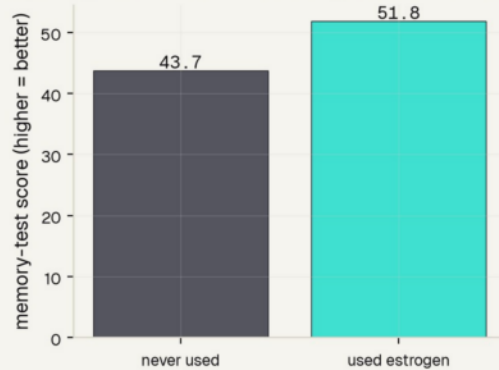


Slide 10 – Results

This opens the Results section, the payoff. Three slides ahead: the tempting raw headline, the honest adjusted result that shrinks it, and a fairness check on the model. Signal that this is where the numbers finally land. Transition: start with the headline anyone would compute first.

The tempting headline

The tempting headline: 'estrogen users think better'



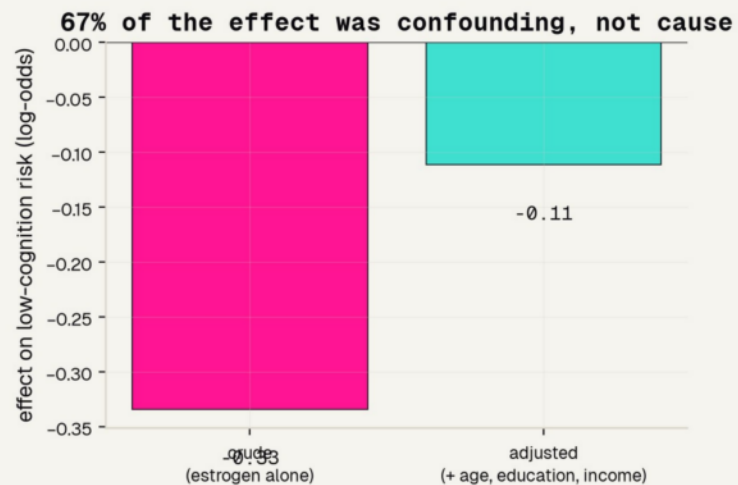
+8.1

points -- the raw gap, users over non-users

51.8 versus 43.7. Taken at face value, the exciting headline writes itself: 'estrogen users think better.' But we never checked whether the two groups were the same kind of people to begin with.

Slide 11 – The tempting headline, exactly as anyone would first compute it. Average the memory score for users versus non-users: 51.8 versus 43.7, an 8-point head start for the estrogen group. If you stopped here, you'd write 'estrogen users think better' -- the same clue the 1990s studies chased. The whole point of the next slide is that we did NOT stop here. Transition: now adjust for who these women were.

Two-thirds of the effect was confounding



67%

of the effect was confounding

The crude effect (-0.33) shrank to just -0.11 once we adjusted for age, education, and income. That missing two-thirds was never the estrogen -- it was that users were healthier and better-off to start with.

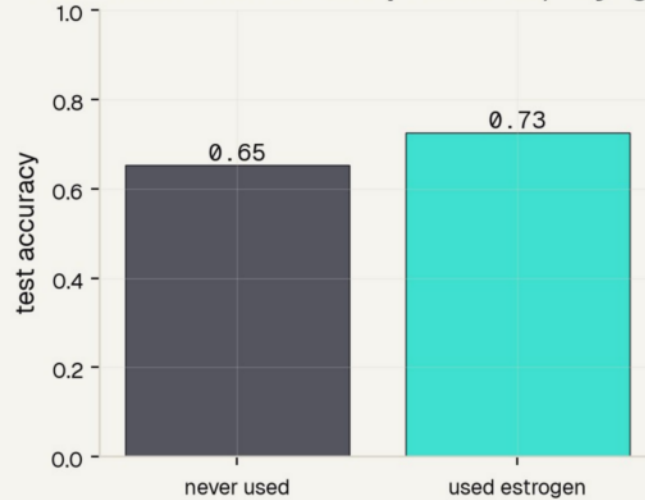
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Slide 12 – Two-thirds of the effect was confounding

The punchline slide. Same data, honest test. Crude effect: about -0.33 (negative = looks protective). Adjusted for age, education, and income: it collapses to -0.11. That's a 67% shrink -- two-thirds of the 'benefit' was never the estrogen, it was that users were healthier and better-off to begin with. This is confounding caught in the act, a miniature of exactly why WHIMS overturned the observational headlines. If they remember one number from this talk, make it 67%. Transition: one more check any people-model needs.

Does the model work for everyone?

How well the model predicts, by group



0.73

test accuracy for estrogen-USERS

0.65

test accuracy for NON-users

A tool that is reliably right for one group and shakier for another can quietly do harm if a clinic trusts it equally. Naming the gap out loud is the honest move.

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Slide 13 – Does the model work for everyone?

A fairness audit, the habit from earlier this week. We trained one model to predict low cognition, then split its accuracy by group: 0.73 for estrogen-users, 0.65 for non-users. A model can look fine overall yet work unequally across groups -- and a clinic that trusts it equally could quietly do more harm to the group it's worse at. The mature move isn't to hide the gap; it's to name it. Transition: and there's something no score in this dataset can see at all.

WHAT THE NUMBERS CAN'T CAPTURE

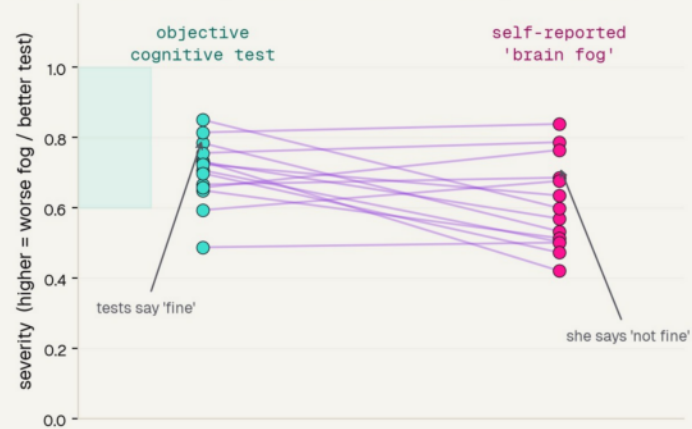
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Slide 14 – What the numbers can't capture

This opens the final section, the honest limits. We step outside the numbers: a test score can say 'fine' while a woman lives with real daily brain fog, and a survey can only hint at cause, never prove it. Framing: good science names what it cannot do. Transition: the tests say fine; she says not fine.

The tests say “fine”; she says “not fine”

The tests can say 'normal' while the patient feels foggy



adapted from Maki 2024, Harvard Health (menopause & brain fog)

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The mismatch

Many women report real, daily brain fog even when the objective test looks perfectly normal [7].

Why it matters

Women's symptoms are too often waved off as stress or emotion instead of investigated [8].

Hold both ideas

The confounding is real AND the patient's account is real evidence -- not noise to dismiss.

Slide 15 – The tests say “fine”; she says “not fine”

Step outside the numbers. Our data is a test SCORE. But many women in menopause report real, daily 'brain fog' even when their objective test looks normal [7] -- the slope lines cross exactly there. And women's symptoms get waved off as stress far too often [8]. The grown-up data-scientist stance holds two true things at once: the confounding is real (the pill may not be the hero), AND the patient's lived report is real evidence, not noise. Transition: so what can this project honestly claim?

A survey can hint; only an experiment proves

WHAT IT CAN DO

Show that a tempting headline is mostly a mirage, and that adjusting for background shrinks the apparent estrogen effect by about 67%.

WHAT IT CANNOT DO

Prove cause. We can only correct for the confounders we actually measured and thought of -- something unrecorded could still be doing the work.

WHAT SETTLES IT

A randomized trial: a coin flip, not the patient, assigns the drug, so the two groups start out identical. That is why WHIMS could overturn 20 years of headlines.

Slide 16 – A survey can hint; only an experiment proves

Be scrupulously honest about scope -- this is what earns trust. CAN DO: show a tempting headline is mostly a mirage and that adjusting shrinks it ~67%. CANNOT DO: prove cause -- we can only correct for confounders we measured and thought of; something unrecorded could still be doing the work. SETTLES IT: a randomized trial, where a coin flip (not the patient) assigns the drug, so the groups start identical. That's why WHIMS could do what twenty years of watching couldn't. Transition: the sources.

References

[1] Tang et al. 1996, Lancet -- observational; estrogen users got Alzheimer's less/later.

[2] Zandi et al. 2002, JAMA (Cache County) -- observational; long-term users lower risk.

[3] Shumaker et al. 2003, JAMA -- WHIMS RCT; hormones DOUBLED dementia risk in 65+.

[4] Rapp et al. 2003, JAMA -- WHIMS RCT; no cognitive benefit, scores trended worse.

[5] Wharton et al. 2009, Maturitas -- healthy-user bias; users start out healthier.

[6] Vandenbroucke 2009, Lancet -- why observational studies and RCTs disagreed.

[7] Maki 2024, Harvard Health -- menopause 'brain fog' vs. normal test scores.

[8] Chen & Shafir 2025, Harvard Health -- the dismissal of women's symptoms.

Slide 17 – References

The paper trail, in order of the story. Top-left: the two observational studies that raised the exciting clue

[1, 2]. Bottom of the left column into the trial: the WHIMS randomized results that overturned it [3, 4].

Right column: the healthy-user-bias explanation [5, 6] and the brain-fog / symptom-dismissal pieces [7, 8].

All were checked against PubMed or the journal page. Transition: one habit to take with you.

The one habit to carry out the door

Suspect the chooser, not the treatment.

When a group that CHOSE a treatment looks healthier, the head start usually belongs to the person, not the pill -- and only a coin-flip experiment can promote a correlation to a cause.

And take the patient's account seriously -- even when the test says 'fine.'

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Slide 18 – The one habit to carry out the door

Close on the transferable lesson, not the estrogen specifics. Suspect the chooser, not the treatment: whenever a group that CHOSE something looks healthier, the head start usually belongs to the person, and only a coin-flip experiment can turn a correlation into a cause. And -- because this is medicine, about real women -- take the patient's account seriously even when the test says 'fine.' That pair, statistical humility plus listening, is the whole point of the project. Thank you.

Thank You

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